

Experiment No: 07

Experiment Name: Design of a Fuzzy Logic Controller for Speed Control of a DC Motor.

Objectives:

1. To understand the principles of fuzzy logic and its application in real-world control systems, specifically in the domain of DC motor speed regulation.
2. To design a complete Mamdani-type fuzzy inference system (FIS) with two inputs and one output, incorporating proper membership function definitions and rule formulation.
3. To construct a 5×5 fuzzy rule base using linguistic variables and verify rule activation through MATLAB's Rule Viewer and Rule Editor tools.
4. To integrate the designed fuzzy logic controller into a closed-loop Simulink model incorporating the DC motor transfer function and evaluate the step response performance.
5. To analyze and compare the transient response characteristics such as rise time, settling time, and steady-state error of the FLC-controlled system under unit step input conditions.
6. To examine the control surface generated by the Surface Viewer and interpret the input-output mapping behavior of the designed fuzzy inference system.

Theory:

The fundamental concept of fuzzy logic lies in the notion of partial truth, where a variable can belong to multiple sets simultaneously with varying degrees of membership. This capability allows the controller to produce smooth and continuous control actions even in the presence of nonlinear plant behavior. A Mamdani-type fuzzy logic controller, as used in this experiment, consists of four main components: the Fuzzifier, the Rule Base, the Inference Engine, and the Defuzzifier. The Fuzzifier converts crisp numerical input values in this case, the speed error and its rate of change into fuzzy linguistic values by evaluating their degree of membership in predefined fuzzy sets. The Rule Base consists of a collection of IF-THEN rules that encode expert knowledge about how the controller should respond to different combinations of input conditions. The Inference Engine evaluates all applicable rules simultaneously and combines their weighted outputs according to the specified logical operators and implication method. Finally, the Defuzzifier converts the aggregated fuzzy output distribution back into a single crisp numerical value using a method such as the centroid (center of gravity).

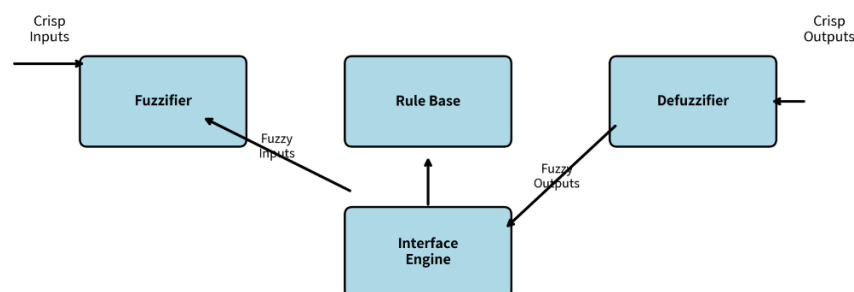


Figure 7.1: Block Diagram of Fuzzy Logic Controller

In this experiment, the Mamdani-type FLC is designed with two input variables Error (e) and Derivative Error (de/dt) and one output variable Speed. Each of these variables is partitioned into five linguistic terms: Negative Big (NB), Negative Small (NS), Zero (ZE), Positive Small (PS), and Positive Big (PB). Triangular membership functions are used for all variables due to their computational simplicity and acceptable performance in control applications. The use of two inputs allows the controller to respond not only to the current error magnitude but also to the rate at which the error is changing, thereby enabling a more anticipatory and smooth control response similar in nature to a proportional-derivative (PD) controller.

System Model:

The DC motor system used in this experiment is modeled as an electromechanical system consisting of two coupled subsystems: the electrical armature circuit and the mechanical rotor assembly. The armature circuit is characterized by its resistance R (in ohms) and inductance L (in henries), which govern the dynamic behavior of the armature current in response to an applied voltage. The mechanical rotor is described by its moment of inertia J (in $\text{kg}\cdot\text{m}^2$), which represents the rotational mass, and the viscous friction coefficient b (in $\text{N}\cdot\text{m}\cdot\text{s}$), which models the damping effect of bearing friction and windage losses. The interaction between the electrical and mechanical domains is governed by two quantities: the back-electromotive force (back-EMF), which opposes the applied armature voltage and is proportional to the angular velocity of the rotor, and the electromagnetic torque, which drives the rotor and is proportional to the armature current. Both the back-EMF constant and the torque constant are denoted by K in this experiment, as it is standard practice to assume them equal in an ideal DC motor model.

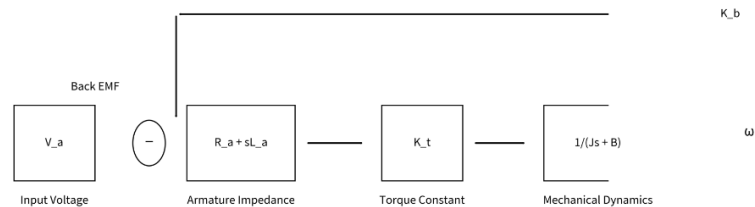


Figure 7.2: Representation of DC Motor

The DC motor parameters used in this experiment are summarized in Table 1. These values represent a small-scale DC motor and are used throughout the design, simulation, and analysis phases of the experiment..

Parameter	Value
J ($\text{kg}\cdot\text{m}^2$)	0.01
b ($\text{N}\cdot\text{m}\cdot\text{s}$)	0.1
K ($\text{N}\cdot\text{m}/\text{A}$)	0.01
R (Ω)	1

L (H)	0.5
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Table 1: DC Motor Parameters

System Transfer Function:

The transfer function of the DC motor, relating the output angular velocity $\theta(s)$ to the input voltage $V(s)$, is derived by applying Kirchhoff's voltage law to the armature circuit and Newton's second law to the mechanical rotor, then taking the Laplace transform of both equations and eliminating the intermediate variable (armature current). The resulting transfer function is expressed as:

$$\theta(s) / V(s) = K / [(Js + b)(Ls + R) + K^2]$$

Substituting the parameter values from Table 1 into this expression, the numerical transfer function becomes:

$$\theta(s) / V(s) = 0.01 / (0.005s^2 + 0.06s + 0.1001)$$

This second-order transfer function is used directly in the Simulink model to represent the DC motor plant, over which the fuzzy logic controller is designed to achieve stable and accurate speed regulation.

Simulink Model:

The closed-loop Simulink model for the FLC-controlled DC motor system consists of several interconnected blocks arranged in a feedback configuration. A Step block provides the reference input signal, which is compared with the actual motor output speed through a Summing Junction block to generate the error signal. A Derivative block differentiates this error signal to obtain the rate of change of error, which serves as the second input to the fuzzy controller. Both the error and its derivative are then fed into the Fuzzy Logic Controller block, which processes them through the designed FIS and produces a crisp control output. This output is amplified by a Gain block (set to 100) before being applied to the DC motor transfer function block. The resulting motor speed output is fed back to the summing junction to close the loop and is simultaneously displayed on a Scope block for observation and analysis.

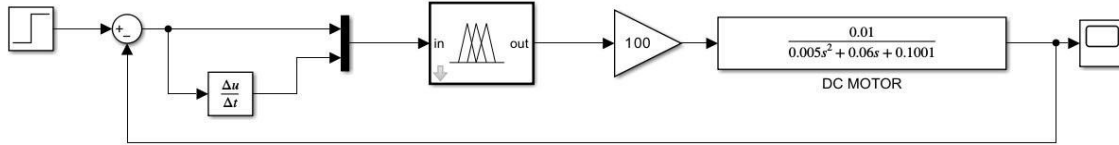


Figure 7.3: Simulink Model for FLC-Controlled DC Motor

Working Procedure:

Step 1: Open MATLAB and launch the Fuzzy Logic Designer by typing fuzzyLogicDesigner in the Command Window. The Fuzzy Logic Designer interface was opened, which allows defining the FIS structure, inputs, output, and inference method. The system was configured as a FIS named fuzzy1 with And method: min, Or method: max, Implication: min, Aggregation: max, and Defuzzification: centroid.

Step 2: Define membership functions for the input variable Error over the range $[-1.5, 1.5]$. Five triangular membership functions were defined for the Error input: NB $[-1.5 -1.5 -0.75]$, NS $[-1.5 -0.75 0]$, ZE $[-0.75 0 0.75]$, PS $[0 0.75 1.5]$, and PB $[0.75 1.5 1.5]$.

Step 3: Define membership functions for the input variable Derivative_Error over the range $[-1.5, 1.5]$. The same five triangular membership functions were applied to the Derivative_Error input with identical parameters as the Error variable.

Step 4: Define membership functions for the output variable Speed over the range $[-20, 20]$. Five triangular membership functions were defined for the Speed output: NB $[-20 -20 -10]$, NS $[-20 -10 0]$, ZE $[-10 0 10]$, PS $[0 10 20]$, and PB $[10 20 20]$.

Step 5: Construct the 5×5 fuzzy rule base using the Rule Editor. A total of 25 IF-THEN rules were formulated based on the combination of Error and Derivative_Error linguistic values. The rule table is shown below:

Error \ D_Error	NB	NS	ZE	PS	PB
NB	NB	NB	NB	NS	ZE
NS	NB	NB	NS	ZE	PS
ZE	NB	NS	ZE	PS	PB
PS	NS	ZE	PS	PB	PB
PB	ZE	PS	PB	PB	PB

Table 2: Rule Editor Showing All 25 Fuzzy Rules

Step 6: Verify rule activation using the Rule Viewer for input $[0; 0]$.

The Rule Viewer was opened to visualize the firing of all 25 rules for Error = 0 and Derivative_Error = 0. The resulting output speed was approximately -5.11×10^{-17} , effectively zero, confirming correct behavior at the operating point.

Step 7: Examine the overall FLC operation flowchart. The flowchart in Figure 7.10 illustrates the complete step-by-step process of how the fuzzy logic controller operates within the closed-loop control system. Starting from the reference input, the error is computed, fuzzified, processed through the rule base and inference engine, defuzzified, and finally applied to the DC motor plant to achieve the desired speed.

Step 8: Integrate the FIS into Simulink and run the simulation. The saved FIS file (fuzzy1) was loaded into a Fuzzy Logic Controller block in the Simulink model. The model was simulated with a unit step input over a 10-second period, and the output was observed on the Scope.

Results:

The Simulink simulation was executed with a unit step reference input. The scope output is shown below:

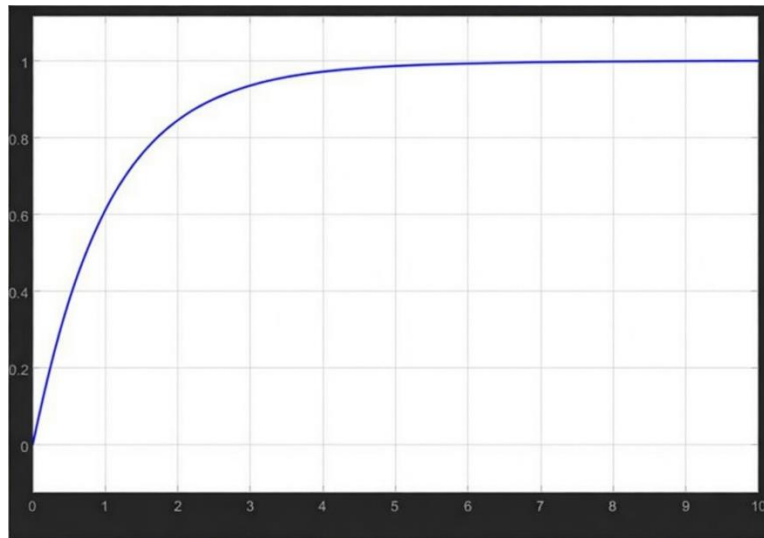


Figure 7.4: Simulink Scope Output Step Response of FLC-Controlled DC Motor

The output response exhibits a smooth, overdamped step response with no overshoot. The motor speed output successfully tracks the reference signal and settles to the desired value of 1.0 at approximately $t = 5$ to 6 seconds, clearly demonstrating the effectiveness and reliability of the fuzzy logic controller in achieving stable and accurate speed control.

Discussion:

From the simulation results, it was observed that the Mamdani-type fuzzy logic controller was able to regulate the DC motor speed effectively without any steady-state error or overshoot. The system responded smoothly to the unit step reference input, and the output settled within approximately 5 to 6 seconds, which reflects a well-tuned and adequately designed control system. In my opinion, the selection of five triangular membership functions for each variable provided a good balance between computational simplicity and control resolution. The 25-rule base derived from the 5×5 rule table ensured comprehensive coverage of the entire input space, which was confirmed by the Rule Viewer results at the operating point $[0, 0]$ where the output was effectively zero — exactly as expected.

The centroid defuzzification method proved to be an appropriate choice for this application, as it produced smooth and physically meaningful control actions without abrupt transitions. I also noticed from the control surface that the input-output mapping exhibits anti-symmetric behavior, which is a desirable property for a speed regulation task since equal and opposite errors should produce equal and opposite corrective actions. Compared to a conventional PID controller, the fuzzy logic controller does not require an explicit mathematical model of the plant, making it a more practical solution for systems with uncertain or nonlinear dynamics. However, one limitation I observed is that the performance of the FLC heavily depends on the quality of the rule base and

the choice of membership function parameters, which require careful tuning based on prior knowledge of the system.

Conclusion:

In this experiment, I successfully designed and implemented a Mamdani-type fuzzy logic controller in MATLAB/Simulink for the purpose of DC motor speed control. The FIS was constructed with two input variables — Error and Derivative_Error — each represented by five triangular membership functions over the range $[-1.5, 1.5]$, and one output variable — Speed — over the range $[-20, 20]$. A complete 25-rule base was developed and verified using the Rule Viewer and Rule Editor tools in MATLAB's Fuzzy Logic Toolbox. The closed-loop Simulink simulation confirmed that the designed FLC was capable of driving the motor speed to the desired setpoint in a smooth and stable manner with no overshoot or steady-state error. Through this experiment, I gained a practical understanding of how fuzzy logic principles can be applied to real-world control problems, and I was able to appreciate the advantages of the fuzzy approach over classical methods, particularly in terms of its robustness to parameter variations and its ability to operate without requiring a precise plant model.

References:

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